

Project Report

Title

TalentOS - AI Recruitment Workspace

Group Members

- Syed Moiz Shahab - Seat No: B23110006164 - Project developer / group member
- Hamza Rizwan - Seat No: B23110006164 - Group member
- Project type: Team project

Introduction

TalentOS is an AI-assisted recruitment platform built with Python and Streamlit. It reduces the manual effort of resume screening by combining PDF text extraction, job-description processing, candidate scoring, ranking, and applicant tracking in one workflow.

Problem Statement

Recruiters often spend significant time reading and comparing CVs manually. This project addresses that problem with a web application that extracts resume content, compares it to job requirements, identifies suitable candidates, and supports the subsequent hiring workflow.

Objectives

- Automate resume and job-description processing.
- Support AI-assisted candidate evaluation.
- Rank candidates based on skill alignment with an open role.
- Provide recruiters with a pipeline, history, and interview workflow.
- Keep a local fallback available when an external AI service is unavailable.

Dataset

The project uses synthetic, non-identifying data for safe demonstration and testing. The dataset contains six demo CV summaries and two job descriptions in the `dataset/` folder. Original readable sample CVs and job descriptions are in `assets/samples/`. No real candidate data is included.

Model and Algorithm

No custom trained or fine-tuned model is used. The application applies PDF text extraction with PyMuPDF and pdfplumber, recognised-skill matching, and deterministic scoring. The algorithm compares skills in a

CV and a job description, identifies matching and missing skills, calculates a bounded 0-100 readiness score, and produces a recommendation. Gemini through LangChain is an optional external analysis path when a valid API key is configured. If it is unavailable, local scoring and workspace retrieval remain available.

System Architecture

The system is organised into Streamlit pages, authentication modules, service classes, SQLAlchemy models, repositories, utility functions, AI modules, sample data, and automated tests. A resume moves through upload, PDF extraction, analysis, ranking, and the ATS pipeline. Sensitive workflow actions are permission checked and audited.

Features Implemented

- User authentication and HR/candidate workflows.
- Resume PDF upload, extraction, cleaning, and email-based candidate de-duplication.
- Job-description entry and optional PDF extraction.
- Candidate summary, skill comparison, readiness score, recommendation, justification, and interview questions.
- Candidate ranking, comparison, and CSV export.
- Applicant pipeline, status tracking, interview scheduling, and candidate history.
- AI assistant with Gemini support when configured and a local retrieval fallback when it is not.
- Demo jobs, synthetic CVs, and automated tests.

Results and Discussion

The application produces extracted CV text, candidate summaries, matching and missing skills, a 0-100 fit score, HR recommendation, interview questions, rankings, and pipeline activity. The system demonstrates a practical way to improve the speed and consistency of early-stage candidate review. Results are decision support only and should be reviewed by a qualified recruiter before a real hiring decision is made.

Conclusion

TalentOS is a working prototype for intelligent recruitment support. It demonstrates end-to-end CV processing and candidate workflow management in a Streamlit application. Future improvements could include OCR for scanned files, persistent cloud object storage, calendar and email integration, and additional fairness and audit controls.

AI Usage Disclosure

OpenAI Codex/ChatGPT was used as a development assistant for code review, debugging, documentation drafting, testing guidance, and UI/workflow improvement suggestions. Gemini is optionally integrated into the application through LangChain for runtime analysis. The accompanying AI usage

declaration provides the complete disclosure.

References

- [Streamlit Documentation](#)
- [LangChain Documentation](#)
- [Google Gemini API Documentation](#)
- [Python, SQLAlchemy, PyMuPDF, and pdfplumber Documentation](#)